The Journal of Open Source Software

tsspiral: An R Package to Create Spiral Version of Time Series Plots

Thiyanga S. Talagala[1]

1 Department of Statistics, Faulty of Applied Sciences, University of Sri Jayewardenepura, Sri Lanka

DOI:

Software

- Review
- Repository
- Archive

Submitted:
Published:

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC-BY).

Summary

tsspiral is an open-source R software for visualising time series on a polar coordinate.

By bringing polar-coordinate-based visualisation into routine time series analysis, tsspiral offers a practical tool for exploring temporal patterns and communicating complex time-dependent structures.

Statement of need

tsspiral beridges these gaps. It aims to overcomes previous limitations, while adding support for

With conventional time series plots it is very difficult to nested temporal structures, compare seasonal patterns across different time scales, identify missing values,particularly in long time series specially when you have daily data. Spiral vrsion of time series plot are useful in addressing these limitations. Creating spiral time series plot often reqires custom codes and careful handing of time series data. Furthermore, when data comes as categorical variables we can not use traditional time series plots to visualise data.

Here is an motivation example. The following graph shows relatil bean price and Dambulla vegetable market in Sri Lanka.

```
library(tidyverse)
library(tsibble)
#install.packages("pak")
#pak::pak("thiyangt/vegetablesSriLanka")
library("vegetablesSriLanka")
vegetables_long_tsibble <- vegetables.srilanka |>
  as_tsibble(key = c(Item, Type, Market),
    index = Date)
beans_dambulla <- vegetables_long_tsibble |>
  filter(Item == "Beans", Type == "Retail", Market == "Dambulla")
ggplot(beans_dambulla, aes(x=Date, y=Price)) + geom_line()
```

State of the field

What has been missing is a reusable tool for creating spiral visualisations of time series data that supports the exploration of temporal patterns across multiple time scales. Compared with conventional time series plotting tools, `tsspiral` adds:

- Spiral visualisations of time series using polar coordinates.
- Alternative representations of temporal patterns, making recurring cycles and seasonal variations easier to explore.

- A framework for visualising temporal hierarchies, from annual and quarterly patterns to monthly, weekly, and daily observations.
- New
- An open-source implementation in R that supports reproducible visualisation workflows.
- A flexible approach that can be incorporated into statistical analysis, research communication, and teaching.

Software design

Each circle is split in to 365. For a leap year circle is divided in to „segments. When visualising quarterly data all dates corresponds to that is colour.

`tsspiral` introduces a new set of `ggplot2` geoms that integrate seamlessly with the `ggplot2` grammar of graphics to create spiral time series plots. This design allows users to generate spiral visualisations within familiar R plotting workflows while retaining the flexibility and composability of `ggplot2`.

```
library(patchwork)
library(tsspiral)
p1 <- ggplot(beans_dambulla, aes(x=Date, y=Price, fill=Price)) + geom_tsspiral() +
p2 <- ggplot(beans_dambulla, aes(x=Date, y=Price, fill=Price)) + geom_tsspiralbar(
p3 <- ggplot(beans_dambulla, aes(x=Date, y=Price, fill=Price)) + geom_tsspiralmiss
p4 <- temporal_hierarchical(
 data = beans_dambulla,
 date = Date,
 value = Price)
(p1|p2)/(p3|p4)
```

Research impact statement

tsspiral expands the ways researchers can explore time series through a circular shape. By representing time series on a polar coordinates, the package provides alternative way to explore temporal patterns, recurring cycles and changes over time. furthermore, this allows to visualise temporal-hierarchcical time series. By aggregating time series under different temporal scales from years to quarters, months, weeks, and days, spiral visualisations can reveal recurring seasonal patterns and relationships between different temporal scales. This allows researchers to explore patterns in the time series within and across years, quarters, months, or other time intervals.

AI usage disclosure

ChatGPt was used to assist with code development and documentation roxygen files writing.

References